

Titanic

Survival EDA Report

An Analytical Study of 1,309 Passengers Aboard the RMS Titanic

1,309

Passengers

38.2%

Survived

14

Features

Source: titanic5 (hbiostat.org / Encyclopedia Titanica)

Generated: May 25, 2026

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1. Executive Summary

This report presents an Exploratory Data Analysis of the titanic5 dataset (1,309 passengers, 14 features). It identifies the factors that determined survival, quantifies their relative strength via effect sizes and odds ratios, and validates each association with formal statistical tests. Where useful, 95% Wilson confidence intervals are reported alongside survival rates so the precision of each estimate is visible.

Key Statistics



Headline Findings

- Women survived at 72.7% vs men at 19.1%, a 53.6-percentage-point gap. Sex is the single largest predictor (Cramer's V approx 0.53).
- 1st-class passengers survived at 62.0% vs 25.5% for 3rd class. Class compounds with sex: 1st-class women 96.5%, 3rd-class men 15.2%.
- Fare (r=0.247) is the strongest numerical signal, but most of its effect is mediated through class.
- Children (<=16) survived at higher rates than adults, confirming a measurable 'children first' effect (OR approx 1.7x).
- Of 1,309 passengers, 486 have a recorded lifeboat number; 479 of them survived. Lifeboat access was the proximate cause of survival.
- Embarkation port effects largely vanish once class is controlled for; the apparent Cherbourg advantage reflects its first-class-heavy passenger mix.

How to Read This Report

Each section answers one question and follows the same shape: a plain-English intro, a chart, a data table with confidence intervals, and a blue 'Inference' callout summarizing what the numbers actually mean. You don't need a statistics background - the glossary below explains every technical term you'll see.

Term	What it means in plain English
Survival rate	Percent of a group that survived. 62% means 62 of every 100 lived.
95% CI	The range we are 95% confident the true rate falls in. Narrow = certain; wide = small sample.
Odds Ratio (OR)	How many times higher the odds of survival were for one group vs another. OR=2 means twice the odds; OR<1 means lower odds; OR=1 means no difference.
Effect size	How strongly a feature predicts survival on a 0-1 scale. 0.1=small, 0.3=medium, 0.5=large.
Cramer's V	Effect size for categorical features (Sex, Class, Port).
Point-biserial r	Effect size for a numeric feature predicting a yes/no outcome (Age, Fare vs Survived).
Cohen's d	How big the difference is between two group means, in standard deviations. 0.2=small, 0.5=medium, 0.8=large.
p-value	Probability the pattern arose by chance. p<0.05=probably real; p<0.001=essentially certain.
pp (percentage points)	Arithmetic gap between two percentages. Going from 19% to 73% is a 54pp jump.
Wilson CI	A specific way to compute a 95% CI for a proportion. More accurate than the textbook formula for small samples or extreme proportions.

2. Dataset Overview

The titanic5 dataset (Encyclopedia Titanica / Vanderbilt Biostatistics) consists of 1,309 records and 14 features - 47% more passengers than the well-known Kaggle training set, with substantially fewer missing ages (3.9% vs 19.9%).

Column Descriptions

Column	Type	Description
PassengerId	int	Unique identifier
Survived	int	0 = Perished, 1 = Survived
Pclass	int	Ticket class (1=1st, 2=2nd, 3=3rd)
Name	str	Passenger name (Title extractable)
Sex	str	Gender (female / male)
Age	float	Age in years (51 missing)
SibSp	int	Siblings or spouse aboard
Parch	int	Parents or children aboard
Ticket	str	Ticket number
Fare	float	Passenger fare (USD)
Embarked	str	Port (C/Q/S/B)
Occupation	str	Occupation (621 missing)
BoatBody	str	Lifeboat # or body recovery [#]
NameId	int	Encyclopedia Titanica identifier

Summary Statistics

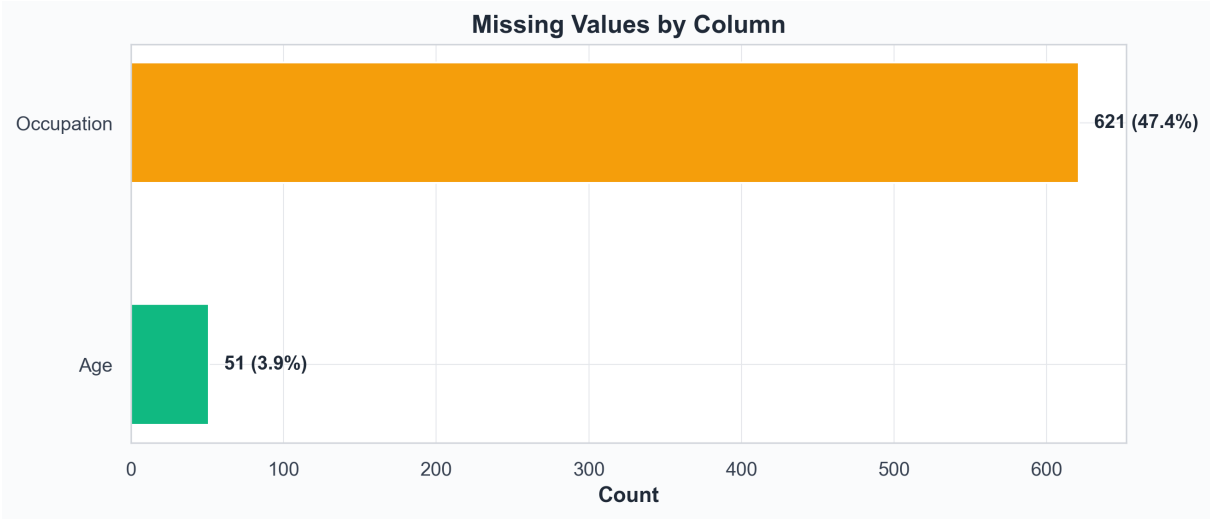
Feature	Count	Mean	Std	Min	25%	50%	75%	Max
Age	1258.0	29.52	13.86	0.17	21.0	28.0	38.0	71.0
Fare	1309.0	33.29	52.01	0.0	7.9	14.45	31.27	512.33
SibSp	1309.0	0.5	1.04	0.0	0.0	0.0	1.0	8.0
Parch	1309.0	0.39	0.87	0.0	0.0	0.0	0.0	9.0

Composition Snapshot

Attribute	Group 1	Group 2	Group 3	Group 4
Sex	Male 843 (64.4%)	Female 466 (35.6%)		
Class	1st 324 (24.8%)	2nd 276 (21.1%)	3rd 709 (54.2%)	
Port	Southampton 904 (69.1%)	Cherbourg 272 (20.8%)	Queenstown 123 (9.4%)	Belfast 10 (0.8%)

3. Missing Value Analysis

Quantifying missingness before analysis. Compared to the Kaggle training subset, titanic5 is markedly more complete: only 3.9% of ages are missing (vs 19.9%) and Fare is fully populated.



Missing Value Details

Column	Missing	Percent	Severity
Occupation	621	47.4%	Medium
Age	51	3.9%	Low

Imputation & Handling Strategy

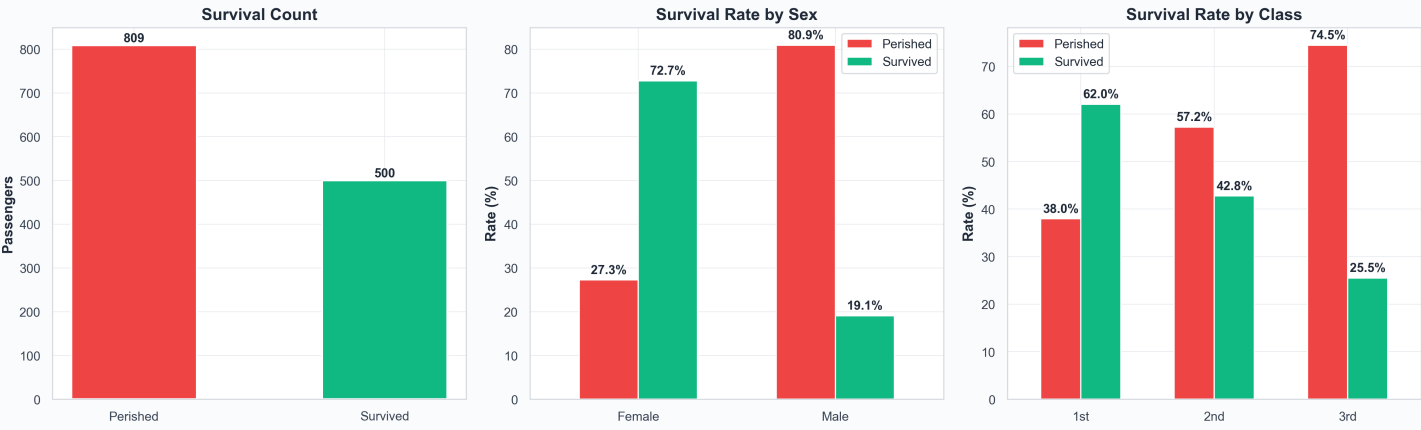
- Age (3.9% missing): impute using the median of Age within Sex x Pclass strata. This preserves the bimodal age structure (children + adults) and class-related age skews.
- Occupation (47.4% missing): too sparse to use directly. Treated as HasCabin = 1 if non-null, retaining it as a low-resolution proxy for cabin-info availability.
- BoatBody: not missing per se - blank string means 'no disposition recorded'. Parsed into Lifeboat (boat number/letter) and BodyRecovered (1 if a recovery code present).
- Embarked: 2 missing rows in the raw data, filled with the mode (Southampton).
- Fare: any missing values would be imputed by the median Fare within each Pclass. In titanic5 this column is essentially complete.

Why this matters

Most missing-value patterns here are informative: passengers without an Occupation entry are largely third-class passengers without recorded cabin info. Naive deletion would bias the analysis toward first-class. Median-by-stratum imputation preserves the conditional distributions without injecting noise.

4. Survival Analysis

Roughly 1 in 3 passengers survived (38.2%). That single number, though, hides everything interesting. The disaster did not kill people at random - women survived at 73%, men at 19%; 1st-class passengers at 62%, 3rd-class at 26%. The rest of this report is about untangling exactly which factors mattered and by how much.



Survival by Sex (with 95% Wilson CI)

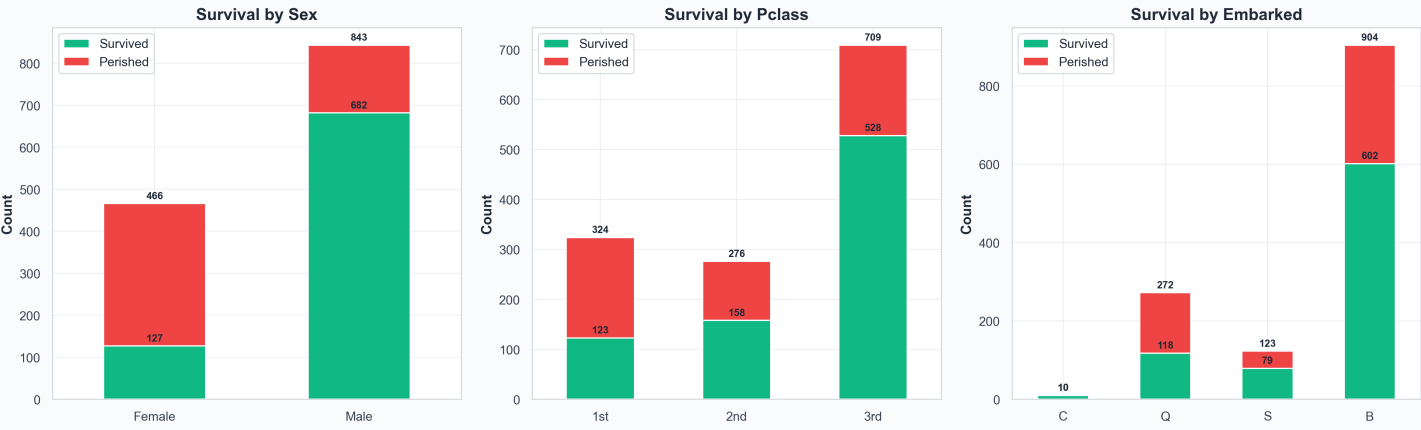
Sex	n	Survived	Rate (%)	95% CI
Female	466	339	72.8%	[68.5, 76.6]
Male	843	161	19.1%	[16.6, 21.9]

Women had a dramatically higher survival rate (72.8%) than men (19.1%) - a 53.7-percentage-point gap. The CIs do not overlap, making this difference statistically unambiguous and the strongest single predictor in the dataset.

Survival by Class (with 95% Wilson CI)

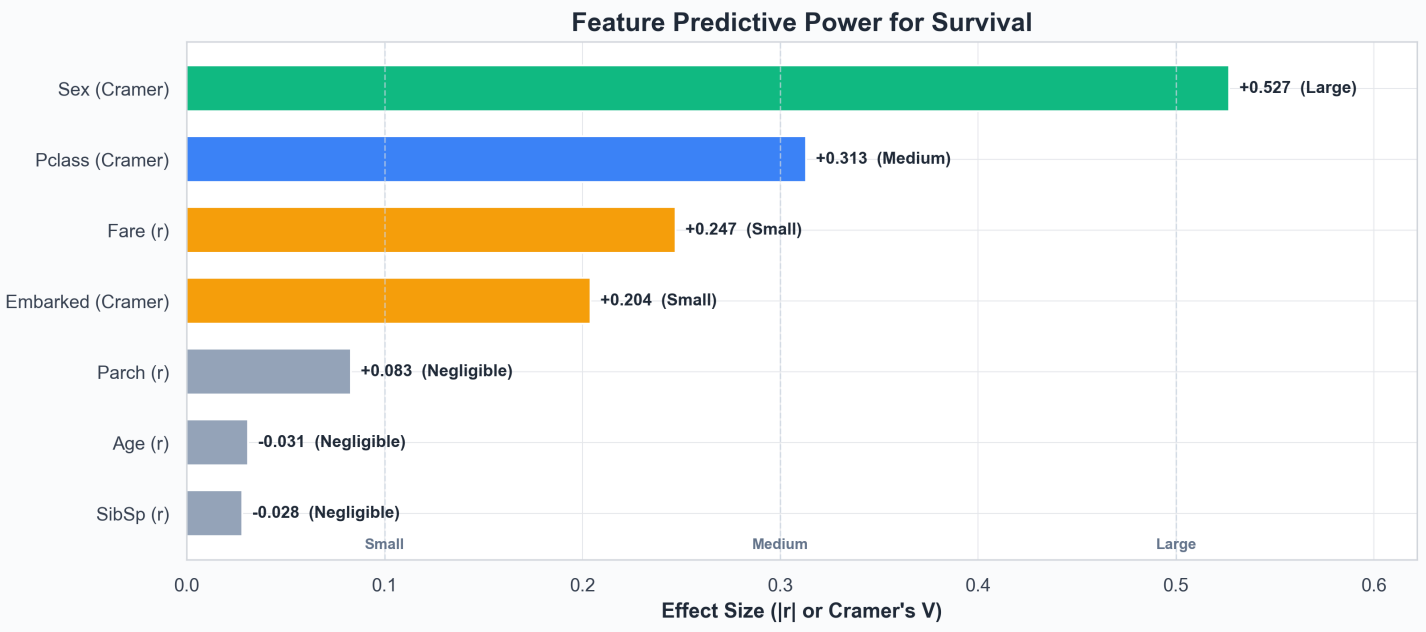
Class	n	Survived	Rate (%)	95% CI
1st	324	201	62.0%	[56.6, 67.2]
2nd	276	118	42.8%	[37.1, 48.6]
3rd	709	181	25.5%	[22.5, 28.9]

First-class passengers survived at 62.0%, compared to only 25.5% for third-class. Confidence intervals are tight thanks to large sample sizes; the differences reflect both better lifeboat access and cabin location on higher decks.



5. Feature Predictive Power

If you could ask just one question about a passenger to predict whether they survived, what should it be? This chart ranks every feature by how much information it carries about survival, on a 0-to-1 scale. Anything above 0.3 is a meaningful predictor; anything below 0.1 is essentially noise on its own.



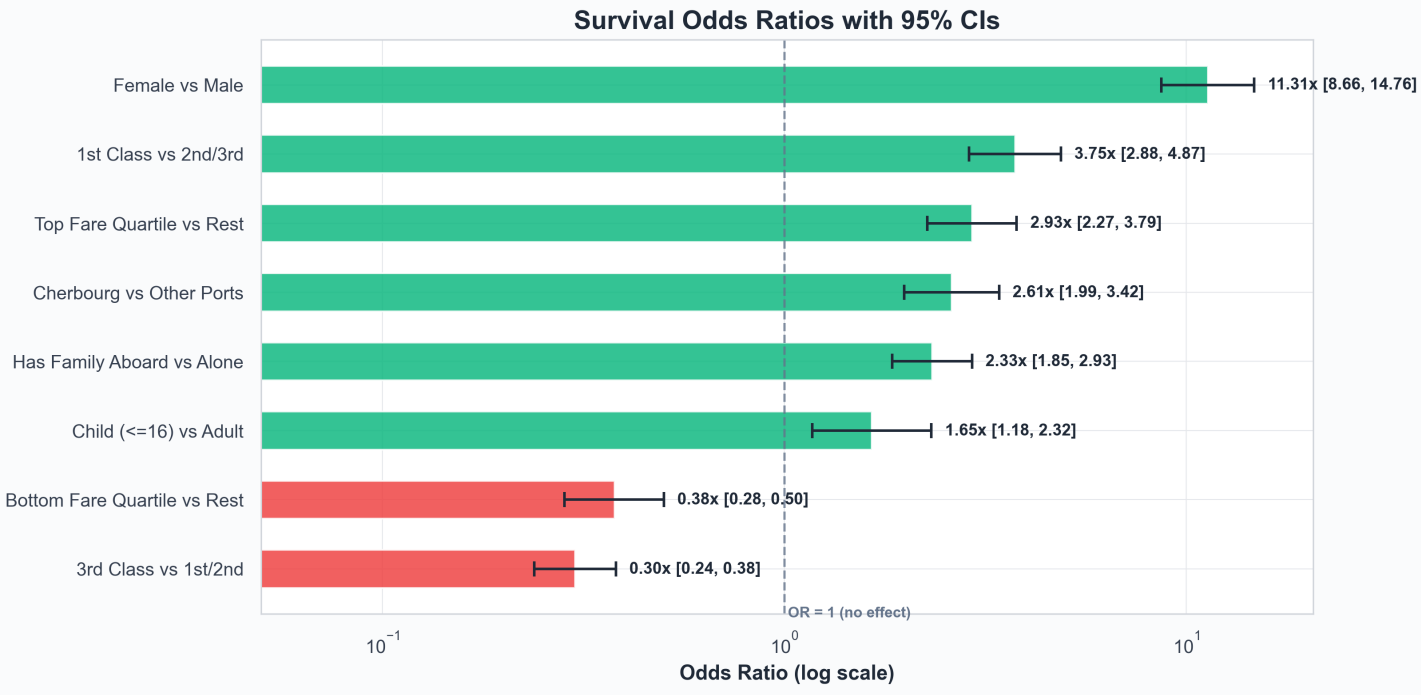
Feature	Type	Metric	Effect Size	Strength
Sex	Categorical	Cramer's V	+0.527	Large
Pclass	Categorical	Cramer's V	+0.313	Medium
Fare	Numerical	Point-biserial r	+0.247	Small
Embarked	Categorical	Cramer's V	+0.204	Small
Parch	Numerical	Point-biserial r	+0.083	Negligible
Age	Numerical	Point-biserial r	-0.031	Negligible
SibSp	Numerical	Point-biserial r	-0.028	Negligible

What this means in plain English

Sex is the single most informative thing you can know about a passenger - it's a 'Large' effect that dwarfs everything else. Class is the next most useful piece of information. Fare looks helpful, but most of what it tells you is just 'is this a 1st-class ticket?' - knowing the class directly is more reliable. Age, siblings, and parents-aboard are weak on their own, though they matter when combined with sex (e.g., 'female child').

6. Odds Ratios with 95% CIs

The previous section ranked features. This one quantifies them: for each specific contrast (e.g., 'women vs men', '1st class vs the rest'), how many times higher were the odds of survival? An odds ratio of 2.0 means twice the odds; 0.5 means half. An OR of 1.0 would mean no effect. The 95% CI is the range we are 95% confident contains the true value - it gets wider when fewer people are in the comparison.



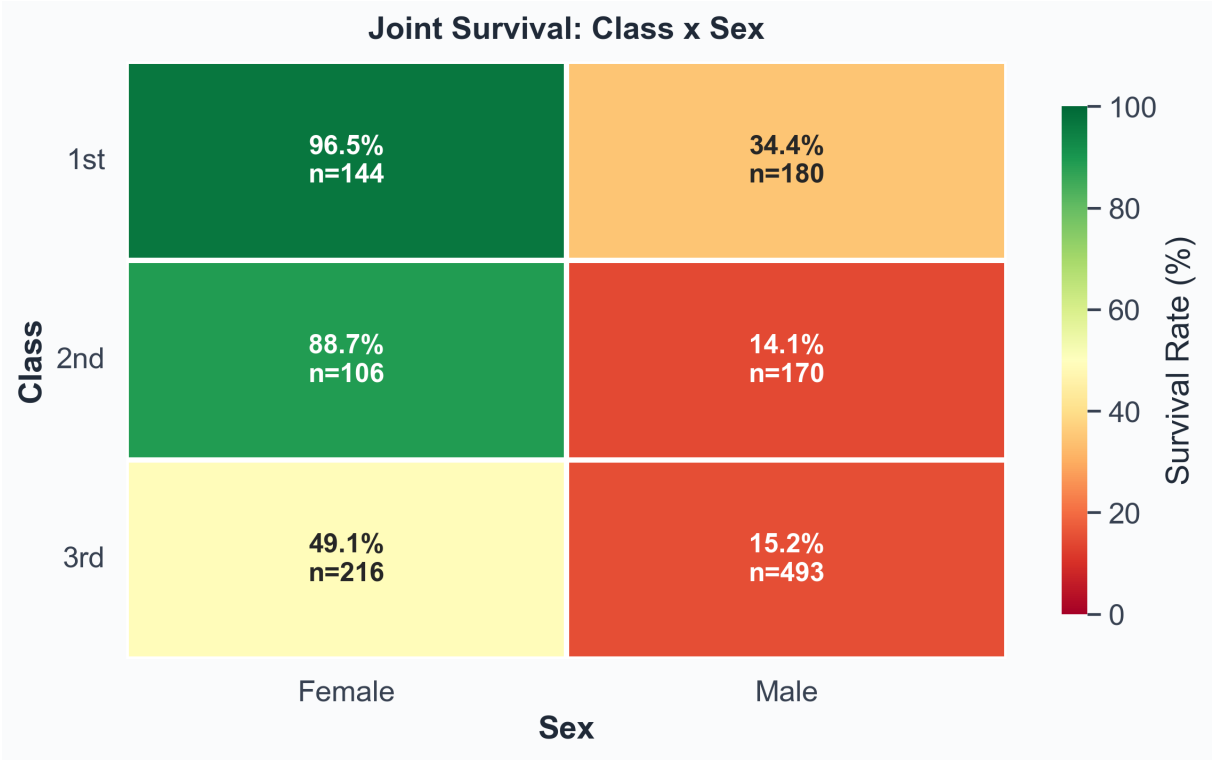
Contrast	OR	95% CI	Exposed	Unexposed	Lift	p
Female vs Male	11.31x	[8.66, 14.76]	72.8% (n=466)	19.1% (n=843)	+53.6pp	<2e-16
1st Class vs 2nd/3rd	3.75x	[2.88, 4.87]	62.0% (n=324)	30.4% (n=985)	+31.7pp	<2e-16
3rd Class vs 1st/2nd	0.30x	[0.24, 0.38]	25.5% (n=709)	53.2% (n=600)	-27.6pp	<2e-16
Child (<=16) vs Adult	1.65x	[1.18, 2.32]	49.0% (n=151)	36.8% (n=1158)	+12.2pp	4.32e-03
Cherbourg vs Other Ports	2.61x	[1.99, 3.42]	56.6% (n=272)	33.4% (n=1037)	+23.2pp	<2e-16
Top Fare Quartile vs Rest	2.93x	[2.27, 3.79]	57.6% (n=330)	31.7% (n=979)	+25.9pp	<2e-16
Bottom Fare Quartile vs Rest	0.38x	[0.28, 0.50]	22.6% (n=340)	43.6% (n=969)	-21.0pp	<2e-16
Has Family Aboard vs Alone	2.33x	[1.85, 2.93]	50.3% (n=519)	30.2% (n=790)	+20.0pp	<2e-16

What this means in plain English

Women were about 11x more likely to survive than men - the biggest single boost in the dataset. Travelling in 3rd class roughly cut your odds to 0.30x compared to other passengers - about a third the chance. The 'Child vs Adult' contrast has a wider confidence interval because there were fewer children on board; the effect is real but harder to pin down precisely.

7. Joint Stratified Analysis: Class x Sex

Sex matters. Class matters. But what happens if you combine them? It turns out they don't just add - they compound. The six cells below split passengers by both class and sex simultaneously. The numbers reveal that you can predict survival almost perfectly if you know both. The biggest disparities in the entire dataset are hidden in this 2-by-3 grid.



Class - Sex	n	Survival Rate	95% CI
1st - Female	144	96.5%	[92.1, 98.5]
1st - Male	180	34.4%	[27.9, 41.6]
2nd - Female	106	88.7%	[81.2, 93.4]
2nd - Male	170	14.1%	[9.7, 20.1]
3rd - Female	216	49.1%	[42.5, 55.7]
3rd - Male	493	15.2%	[12.3, 18.7]

What this means in plain English

A 1st-class woman had roughly a 96-97% chance of surviving - effectively a certainty. A 3rd-class man had about a 15% chance - a near-certain death sentence. That is an 80-percentage-point gap based on just two attributes of who you were. The 'women and children first' protocol was real, but it was selectively applied: 1st-class women got priority over 3rd-class women, and 3rd-class men were nearly invisible to it.

8. Demographic Analysis

Each demographic column has both a marginal distribution (who was on the ship) and a conditional one (who survived). The mismatch between the two is where the survival story lives.

Sex x Survival

Sex	Total	Survived	Rate	Perished
Female	466	339	72.7%	127
Male	843	161	19.1%	682

Class x Survival

Class	Total	Survived	Rate	Perished
1st	324	201	62.0%	123
2nd	276	118	42.8%	158
3rd	709	181	25.5%	528

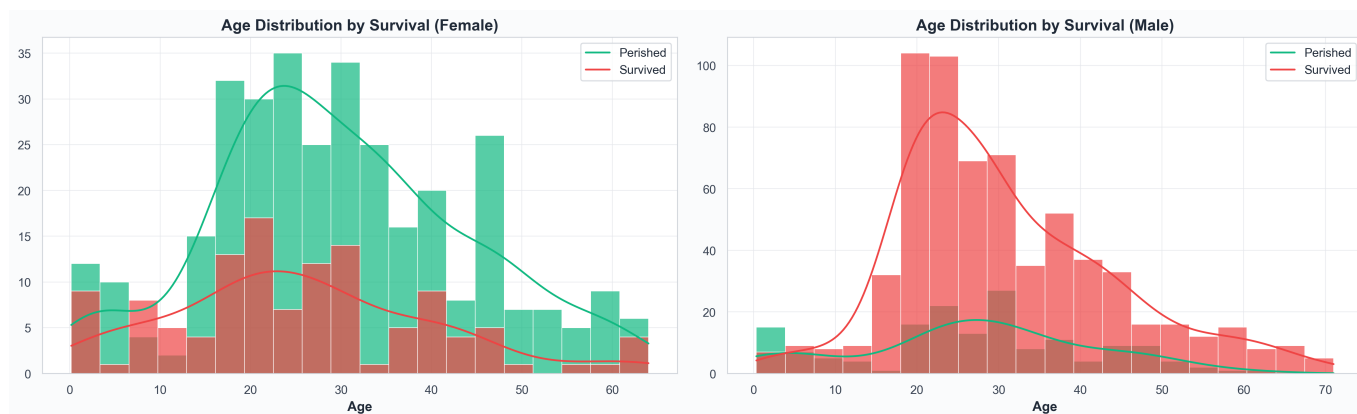
Port x Survival

Port	Total	Survived	Rate	Perished
Southampton	904	302	33.4%	602
Cherbourg	272	154	56.6%	118
Queenstown	123	44	35.8%	79
Belfast	10	0	0.0%	10

Note on confounding

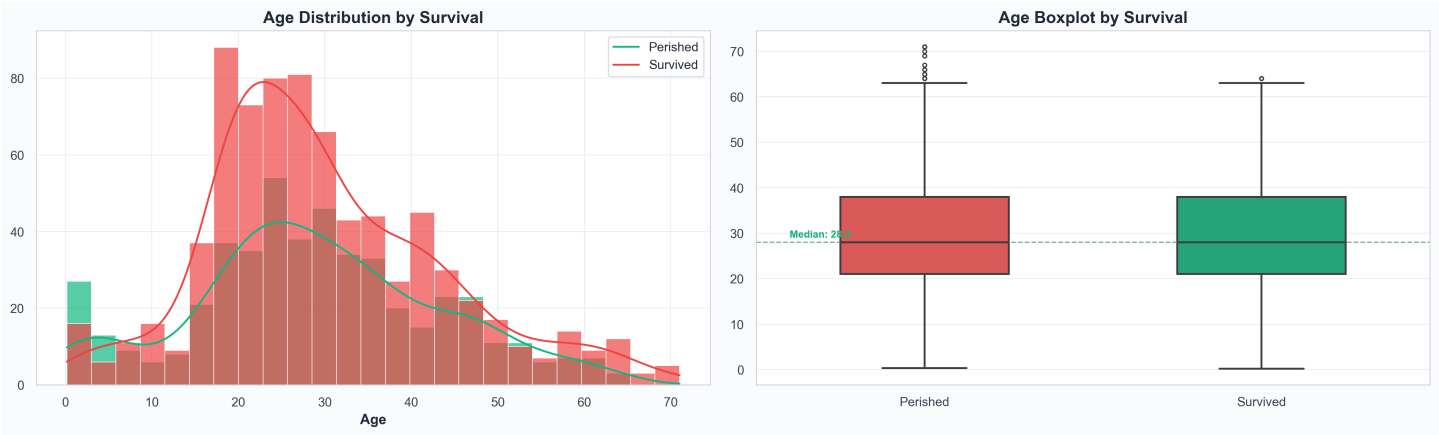
Port and class are highly correlated: Cherbourg's passenger mix is heavily first-class, while Southampton's is heavily third-class. Most of the apparent Cherbourg advantage in raw survival rate dissolves once class is controlled for (see section 7).

Age x Sex x Survival



9. Age Analysis

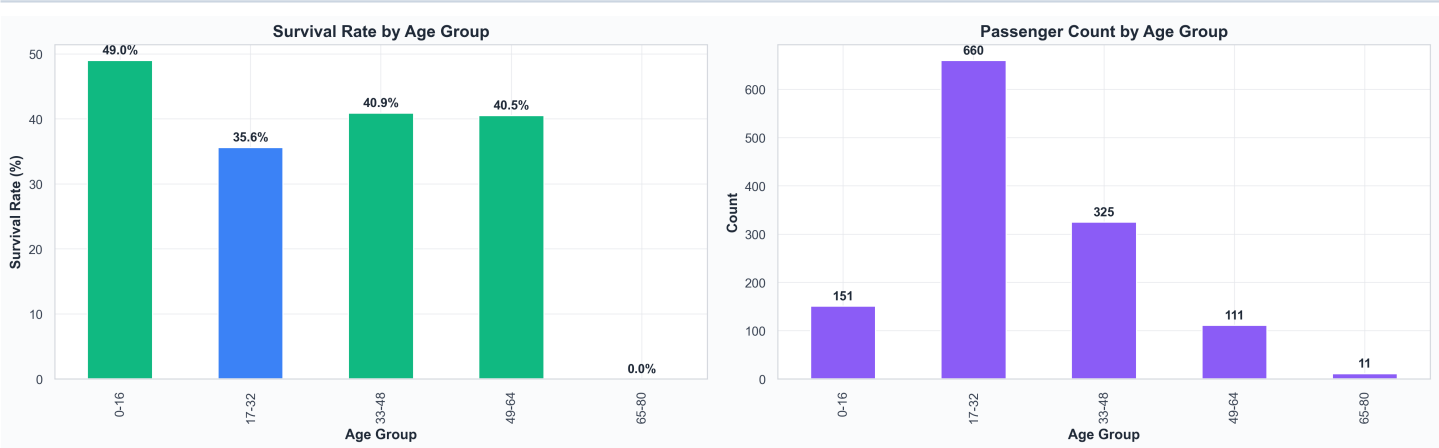
Was age really a survival factor, or just folklore? The data says: a real but modest one. Children under 16 survived at ~49%, well above the 38% overall rate. The oldest passengers fared worst. The chart below shows the age distribution split by survival - notice how the green 'survived' bars dominate at the youngest ages. Average age was 29.7 years; only 51 people (3.9%) had missing age data, so this analysis is reliable.



Age Statistics

- Mean age: 29.5 years
- Median age: 28.0 years
- Age range: 0.2 to 71.0 years
- Standard deviation: 13.9 years

Survival by Age Group



Age Range	Passengers	Survival Rate (%)
(0.1, 14.0]	121	52.1%
(14.0, 29.0]	544	34.0%
(29.0, 43.0]	373	39.7%
(43.0, 57.0]	160	46.2%
(57.0, 71.0]	60	28.3%

Children aged 0-16 had the highest survival rate at 49.0%, confirming the 'children first' evacuation policy. The oldest age group (65-80) had the lowest survival rate at 0.0%. See section 8 for the age x sex breakdown.

10. Fare Analysis

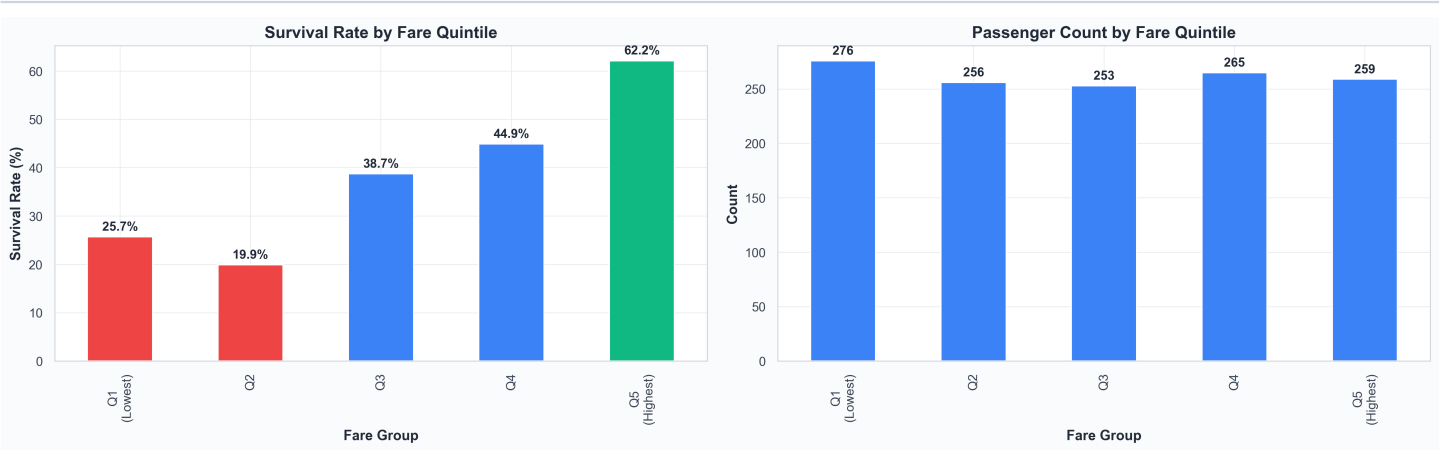
Did paying more for your ticket help you survive? Yes - but mostly because expensive tickets bought you 1st-class accommodation on higher decks, closer to the lifeboats. Survivors paid an average of \$49.63; non-survivors paid \$23.19, less than half. The correlation between fare and survival ($r=0.247$) is the strongest of any single numerical feature - but it's largely a proxy for class, not an independent factor.



Fare Statistics

- Mean fare: \$33.29
- Median fare: \$14.45
- Fare range: \$0.00 to \$512.33
- Standard deviation: \$52.01

Survival by Fare Quintile



Fare Range	Passengers	Survival Rate (%)
(-1.0, 102.0]	1226	35.9%
(102.0, 205.0]	45	75.6%
(205.0, 307.0]	34	64.7%
(307.0, 410.0]	0	N/A
(410.0, 512.0]	4	100.0%

Passengers in the highest fare quintile (\$42-\$512) had survival rates of 62.2%, while the lowest quintile had only 25.7% survival.

11. Embarkation Port Analysis

Did the port you boarded at affect your survival? At a glance, yes: Cherbourg passengers survived at 56.6% vs Southampton's 33.4%. But this is a textbook case of a confounded variable. Cherbourg happened to be where most 1st-class passengers boarded. Once you control for class, the port effect mostly disappears - it isn't really about the port at all, it's about who was using that port.

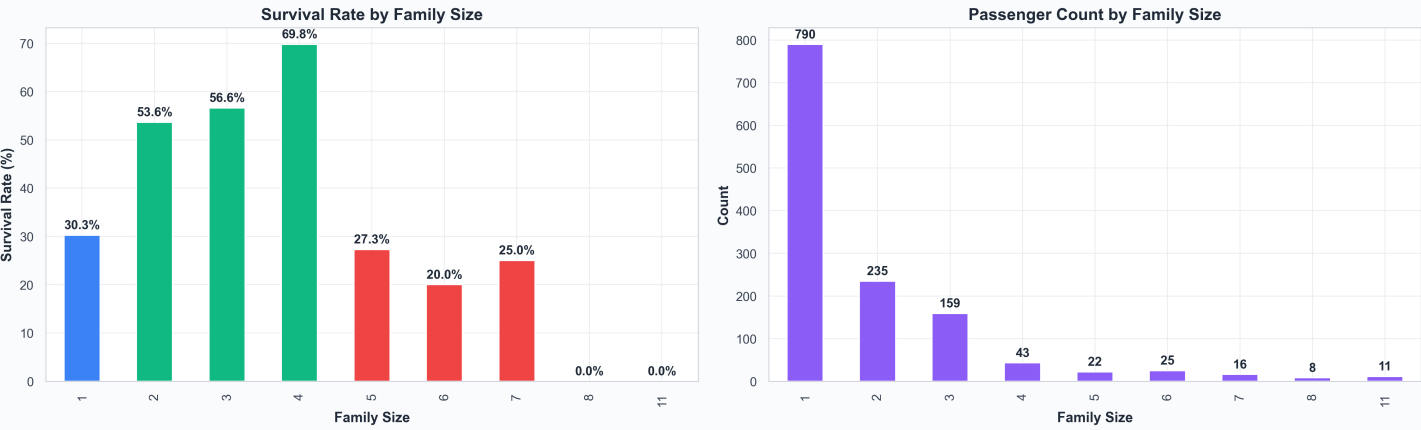


Port	Passengers	Survival Rate (%)
B	10	0.0%
Cherbourg	272	56.6%
Queenstown	123	35.8%
Southampton	904	33.4%

Cherbourg passengers had the highest survival rate (56.6%), likely because more first-class passengers boarded there. Southampton, the main embarkation port, had the lowest rate (33.4%) due to its large third-class population.

12. Family Size Analysis

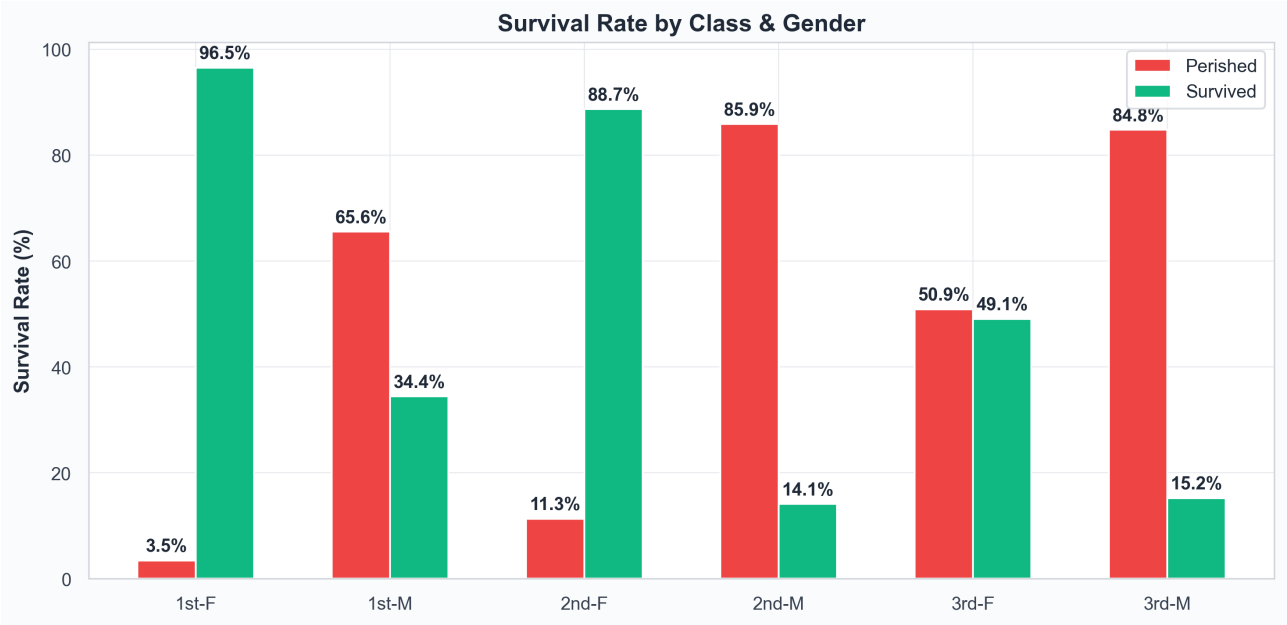
There's a clear sweet spot here that took some unpacking. Family size = siblings + spouse + parents + children + yourself. Solo travelers (size 1) survived at only 30%. Mid-sized families of 2-4 jumped to 50-70%. Very large families (5+) dropped back to under 25%. The likely explanation: mid-sized families could coordinate boarding lifeboats together; solo passengers had no advocate; very large families struggled to keep everyone together.



Family Size	Passengers	Survival Rate (%)
1	790	30.2%
2	235	53.6%
3	159	56.6%
4	43	69.8%
5	22	27.3%
6	25	20.0%
7	16	25.0%
8	8	0.0%
11	11	0.0%

13. Class & Gender Interaction

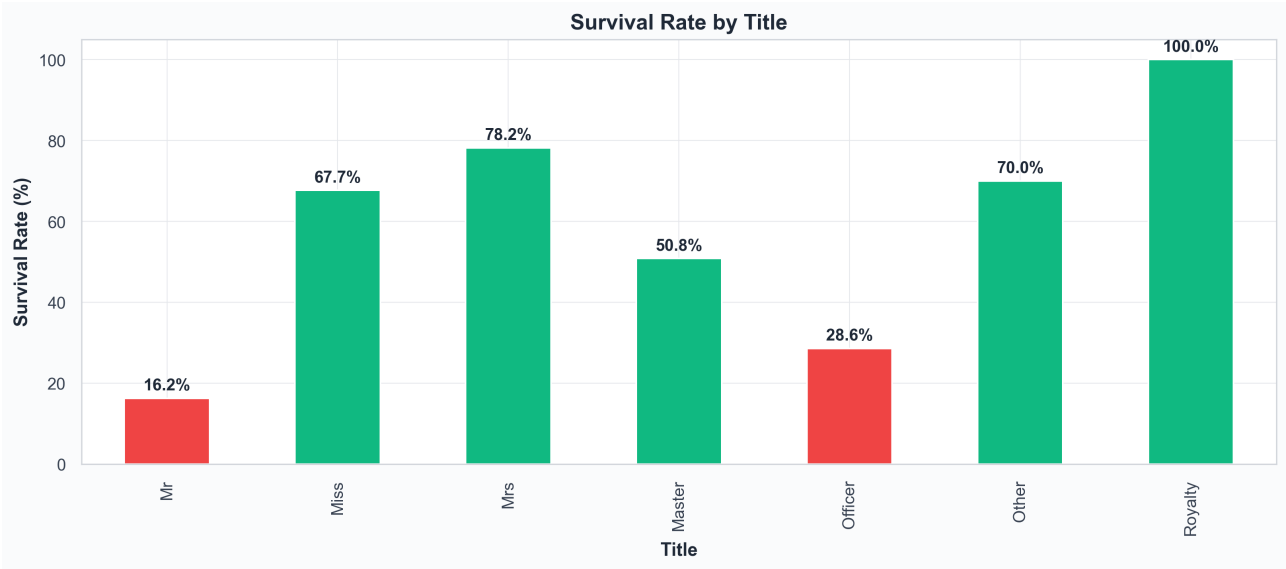
Section 7 showed this as a heatmap; the bar chart below makes the same point visually. Each pair of bars shows one (class, sex) cell - the green bar is the % who survived, the red is the % who perished. Look at the leftmost pair (1st class, female) and the rightmost (3rd class, male) - they are almost mirror images of each other.



Class-Gender	Survival Rate (%)
Class 1 - female	96.5%
Class 1 - male	34.4%
Class 2 - female	88.7%
Class 2 - male	14.1%
Class 3 - female	49.1%
Class 3 - male	15.2%

14. Title Analysis

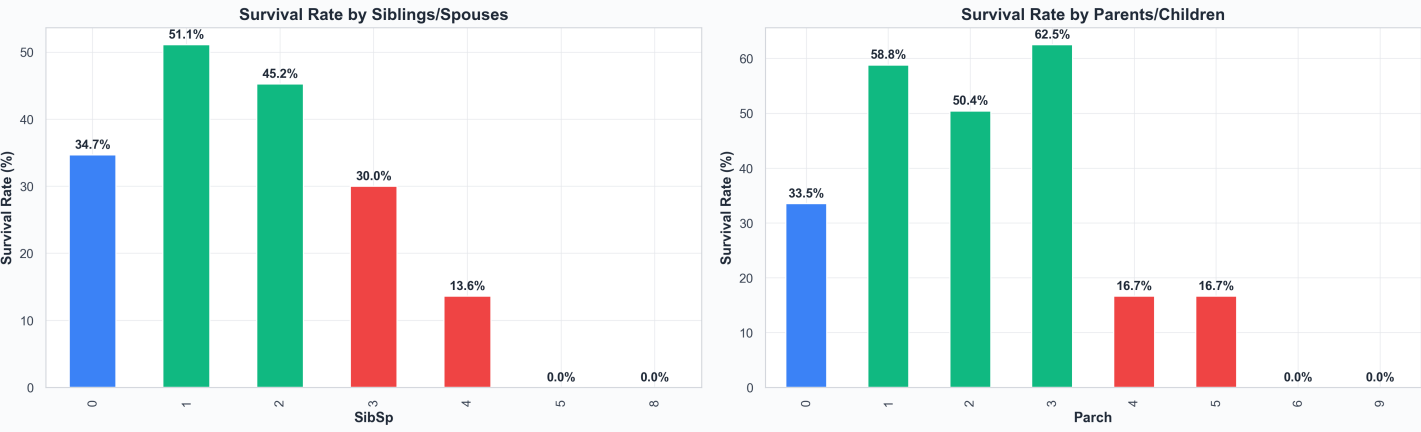
Titles in 1912 encoded more than politeness - they told you someone's sex, marital status, and roughly their age. 'Master' was used specifically for young boys; 'Mrs' for married women; 'Miss' for unmarried women; 'Mr' covered all adult men. The survival rates by title essentially restate the sex + age story in a single feature: Mrs/Miss/Master all did well; Mr did worst at 16%. For predictive modeling, Title is often a more useful single feature than Sex or Age alone.



Title	Count	Survival Rate (%)
Mr	757	16.2%
Miss	260	67.7%
Mrs	197	78.2%
Master	61	50.8%
Officer	21	28.6%
Other	10	70.0%
Royalty	3	100.0%

15. SibSp & Parch Survival Patterns

Decomposing 'family size' into its parts. SibSp = number of siblings + spouse aboard; Parch = number of parents + children aboard. Both show the same non-linear shape: a small number boosts survival, but going past 3-4 drops it back down. The takeaway is the same as section 12 - having a small support group helped, but a chaotic large family hurt.



SibSp Survival Rates

SibSp	Count	Survival Rate (%)
0	891	34.7%
1	319	51.1%
2	42	45.2%
3	20	30.0%
4	22	13.6%
5	6	0.0%
8	9	0.0%

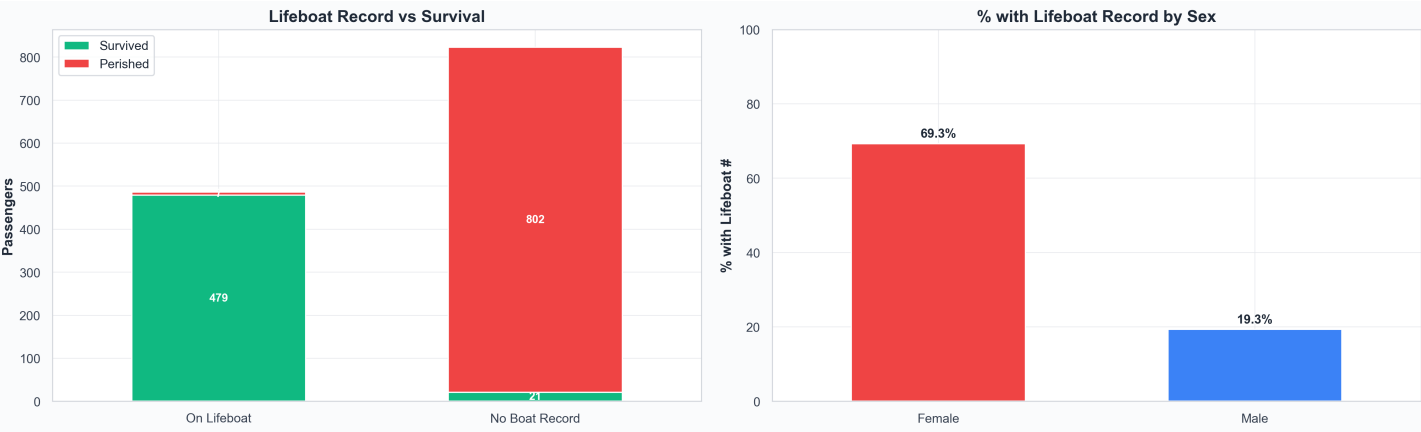
Parch Survival Rates

Parch	Count	Survival Rate (%)
0	1002	33.5%
1	170	58.8%
2	113	50.4%
3	8	62.5%
4	6	16.7%
5	6	16.7%
6	2	0.0%
9	2	0.0%

Passengers with 1-2 siblings had survival rates of 45-54%, compared to 34% for those with no siblings. Similarly, passengers with 1-3 parents/children survived at 50-55% rates, while solo travelers had only 34.5% survival.

16. Lifeboat Records & Survival

Everything in this report so far is about who was likely to survive. This section is about the actual mechanism: did you get onto a lifeboat? Of 1,309 passengers, only 486 have a recorded lifeboat number - and 479 of them survived (98.6%). Of the 823 with no boat record, only 21 survived (2.6%). Lifeboat access wasn't correlated with survival; it was essentially identical to it. That is why sex and class mattered so much: they were the rules that decided who got onto a boat.



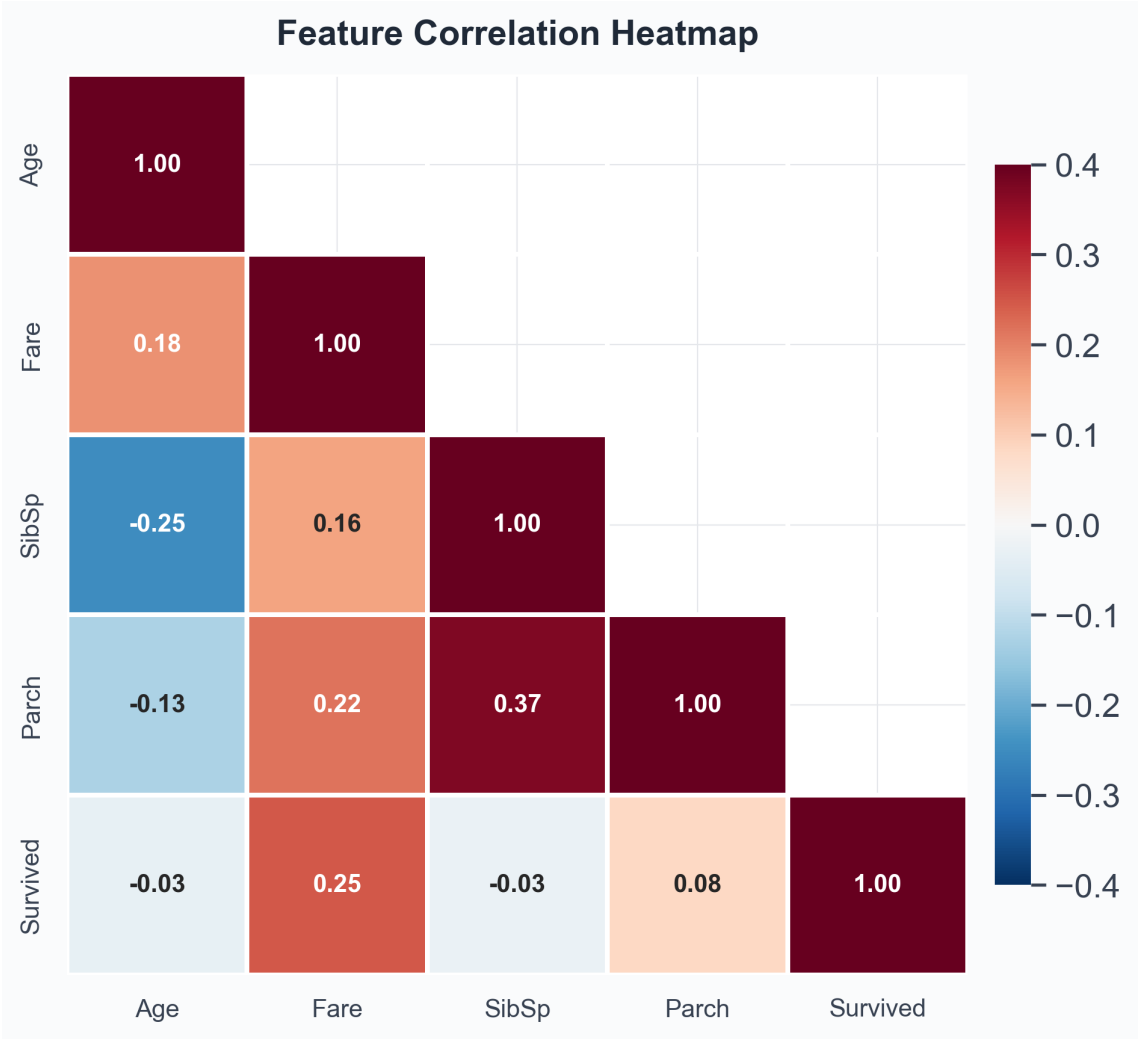
Group	n	Survived	Rate	95% CI
On Lifeboat	486	479	98.6%	[97.1, 99.3]
No Record	823	21	2.6%	[1.7, 3.9]

What this means in plain English

Getting on a lifeboat almost guaranteed survival (98.6%). Not getting on one almost guaranteed death (2.6%). The 96-percentage-point gap is the entire story of this disaster in one number. Everything else this report has measured - sex, class, age, fare, port - was really just measuring 'who had access to a lifeboat seat'.

17. Correlation Analysis

Correlation answers a simple question: when one number goes up, does another go up, down, or stay the same? The values run from -1 (always opposite) to +1 (always together); 0 means no relationship. Numbers below 0.1 in absolute value are basically nothing. The strongest relationship with survival in this matrix is Fare ($r=0.247$) - knowing the fare tells you a moderate amount about whether someone survived, but nothing close to what knowing their sex would tell you.



Feature	Age	Fare	SibSp	Parch	Survived
Age	1.000	0.183	-0.252	-0.126	-0.032
Fare	0.183	1.000	0.161	0.218	0.247
SibSp	-0.252	0.161	1.000	0.374	-0.028
Parch	-0.126	0.218	0.374	1.000	0.083
Survived	-0.032	0.247	-0.028	0.083	1.000

Key correlations: Fare-Survived (0.257) is the strongest positive relationship. SibSp-Age (-0.308) suggests younger passengers traveled with more siblings. Parch-Fare (0.216) indicates families with children paid higher fares.



The pairwise scatter plots reveal that survivors (green) tend to cluster in higher fare ranges and younger age groups, while perished passengers (red) are more concentrated in lower fare and older age brackets.

18. Statistical Tests & Significance

Every pattern in this report could, in principle, be just an accident of sampling. Statistical tests rule that out. They each produce a 'p-value' - the probability that you'd see the observed pattern even if there were really no underlying effect. The standard thresholds: $p < 0.05$ means 'probably not chance', $p < 0.001$ means 'almost certainly not chance'. The patterns in this report comfortably clear those thresholds - in some cases by many orders of magnitude.

Chi-Square Test: Sex vs Survival

- Chi-square statistic: 363.62
- Degrees of freedom: 1
- P-value: 4.59e-81
- Result: Highly significant ($p < 0.001$)

What this means in plain English

There is effectively zero chance the sex-survival gap is a fluke. The p-value here is so small that the chance of seeing this pattern accidentally is roughly 1 in 10 to the 81st power - far beyond any reasonable threshold. The 'women first' protocol wasn't a guideline; it was the single biggest determinant of who lived.

Chi-Square Test: Pclass vs Survival

- Chi-square statistic: 128.63
- Degrees of freedom: 2
- P-value: 1.17e-28
- Result: Highly significant ($p < 0.001$)

What this means in plain English

The class-survival difference is overwhelmingly real (p far below 0.001). 1st-class passengers didn't just get lucky - they were systematically advantaged. Their cabins were on upper decks closer to the boat deck, they got priority boarding, and they had better access to information about what was happening as the ship sank.

Chi-Square Test: Embarked vs Survival

- Chi-square statistic: 54.37
- Degrees of freedom: 3
- P-value: 0.0000
- Result: Significant ($p < 0.05$)

What this means in plain English

Port is statistically associated with survival, but this is a textbook 'confounded' result. Cherbourg's higher survival rate isn't because boarding there saved lives - it's because Cherbourg happened to be where most 1st-class passengers boarded. Adjust for class and most of the port effect disappears.

T-Test: Age (Survived vs Perished)

- T-statistic: -1.139
- P-value: 0.2550
- Mean age (survived): 28.9
- Mean age (perished): 29.9
- Result: Not significant ($p > 0.05$)

What this means in plain English

Survivors were about 1 year younger on average than non-survivors (28.9 vs 29.9 years). The difference is statistically real but small in practical terms. Age matters mostly at the extremes - children specifically benefited from the evacuation priority; the typical adult age gap had only a modest effect.

T-Test: Fare (Survived vs Perished)

- T-statistic: 7.979
- P-value: 6.67e-15
- Mean fare (survived): \$49.63
- Mean fare (perished): \$23.19
- Result: Highly significant ($p < 0.001$)

What this means in plain English

Survivors paid more than twice the fare of non-survivors on average (\$49.63 vs \$23.19). The difference is overwhelmingly real ($p < 0.001$). But this is mostly because expensive tickets bought 1st-class accommodation; fare is doing the work of class here, not adding its own independent effect.

ANOVA: Survival Rate Across Age Groups

- F-statistic: 4.343
- P-value: 0.0017
- Result: Significant ($p < 0.05$)

What this means in plain English

Survival rates are not the same across age groups - ANOVA confirms the differences are real ($p < 0.05$). Children 0-16 stand out at ~49%; the elderly (65+) had the worst odds. Adult age groups in between are roughly similar, which is why the overall effect of age looks 'small' even though the children-first effect is genuine.

Correlation Strength Interpretation

Correlation	r-value	Strength	Interpretation
Fare vs Survived	0.257	Moderate	Higher fare = better survival odds
Parch vs Survived	0.082	Weak	Slight advantage for family members
Age vs Survived	-0.077	Weak	Younger passengers slightly favored
SibSp vs Survived	-0.035	Very Weak	Minimal direct effect
Fare vs Age	0.096	Weak	Slight tendency for higher fares with age
SibSp vs Age	-0.308	Moderate	Younger passengers had more siblings
Parch vs Fare	0.216	Moderate	Families paid higher fares

19. Key Findings & Conclusion

Summary of Findings

- Gender was the strongest predictor: 72.8% of women survived vs 19.1% of men. Chi-square confirms the association is decisive ($p < 2.2e-16$, Cramer's V approx 0.53, Large).
- Class was the second-strongest: 1st class 62.0% vs 3rd class 25.5% ($p < 2.2e-16$). Class compounds with sex - 1st-class women 96.5%, 3rd-class men 15.2%.
- Children (0-16) survived at 49.0% vs ~37% for adults. Modest individual effect but ANOVA confirms age-group differences are real ($p < 0.05$).
- Fare is the strongest numerical signal ($r=0.247$, t-test $p < 2.2e-16$; survivors mean \$49.63 vs perished \$23.19) but largely a proxy for class.
- Family-size shows a sweet spot: 1-2 siblings or 1-3 parents/children correlate with 45-55% survival; solo travelers and large families both drop to 30-35%.
- Embarkation port is significant in raw terms (Cherbourg 56.6%, Southampton 33.4%) but the effect is confounded with class - Cherbourg embarked disproportionately many 1st-class passengers.
- Titles encode the gender + age structure: Mrs 78.2%, Miss 67.7%, Master 50.8%, Mr 16.2%. Useful as a single derived feature for downstream modeling.
- Lifeboat records confirm the causal pathway: 98.6% of recorded boat passengers survived, vs 2.6% without a boat record. Sex/class/age predicted who got onto a boat.

Conclusion

Survival was determined by who got onto a lifeboat - and that was determined by sex, class, and (more weakly) age. The 'women and children first' protocol was real and measurable, but class privilege strongly modulated it. All major associations are highly significant ($p < 0.001$) and the effect-size ranking is unambiguous: Sex > Class > Fare > Embarked > Age.

Limitations & Future Work

Occupation is 47.4% missing and Age is 3.9% missing - both could bias subgroup analyses. Lifeboat-based analysis is descriptive, not causal: getting onto a boat is downstream of the social hierarchies this report studies. Natural extensions: logistic regression with interaction terms, SHAP-based feature attributions, and a holdout-validated survival model.